

PRASIDDDHI DUMRE

☎ 981-4820842 ✉ dumreprasiddhi@gmail.com 📍 Budhanilkantha, Kathmandu ➡ <https://github.com/Prasiddhi16>

Personal statement

Enthusiastic Computer Engineering student at Kathmandu University with hands-on research experience at the Digital Learning Research Lab (DRL-Lab). Passionate about building software solutions, contributing to open-source, and driving collaborative projects. Proven leadership in team environments with a strong foundation in full-stack development, machine learning, and systems programming.

Skills

- **Programming & Web Development:** Python, TypeScript, JavaScript, C/C++, HTML/CSS, SQL
- **Libraries & Frameworks:** OpenCV, MediaPipe, REST APIs, Django, NumPy, Scikit-learn
- **Visualization & UI:** Interactive Visualization, Dashboard Design
- **Databases:** MySQL database management
- **Verbal and written Communication**
- **Data & AI/ML:** Computer Vision, Image Classification, Predictive Analytics, Data Structures, Optimization
- **Algorithms & Problem Solving:** Graph Algorithms (Dijkstra, BFS, DFS), Adjacency List Modeling
- **Tools & Platforms:** Git, GitHub, Makefile, VS Code
- **Technical writing**
- **Research proficiency**

Experience

Jan 2024 – Current

RESEARCH INTERN

Digital Learning Research Lab (DRL-Lab), Kathmandu University

- Contributing to research on digital learning technologies and educational software systems
- Designing and implementing experimental prototypes to support ongoing lab research initiatives
- Collaborating with faculty and fellow researchers on data collection, analysis, and documentation
- Gaining experience in academic writing, literature reviews, and presenting findings to the team

Education

2024 – Expected graduation 2028

BACHELOR OF ENGINEERING COMPUTER ENGINEERING CANDIDATE

Kathmandu University, Dhulikhel

2022 – 2024

HIGHER SECONDARY EDUCATION SCIENCE

Kathmandu Model College, Bagbazar

Teamwork Experience

Worked with a cross-functional team to successfully deliver a semester-long engineering project. Organized and participated in workshops, code reviews, and knowledge-sharing sessions for junior students.

Projects

- **Artha-AI :** Personal Finance Management System, HTML, JavaScript, AI/ML. Built an AI-powered finance app enabling users to track income, expenses, budgets, and goals with predictive analytics. Integrated intelligent alerting and receipt scanning to automate expense categorization. Designed an intuitive dashboard with visual analytics for real-time financial insights.
- **Waste-Sorting-Assistant :** AI Mobile App Model, TypeScript, Computer Vision. Developed an AI-powered mobile application that classifies waste materials from camera images in real time. Guides users on correct sorting and disposal, promoting smarter recycling and sustainability practices. Combined image classification models with a clean mobile UI for a seamless user experience.
- **Bloom :** Gesture Recognition & Visual Effect Rendering, Python, OpenCV, MediaPipe. Created a real-time hand gesture recognition system that maps gestures to dynamic visual effects. Leveraged computer vision pipelines to track hand

landmarks at high frame rates with low latency. Demonstrated creative application of CV techniques — blending research curiosity with interactive design.

- **Smart-Bus-Route-Optimizer** : C, C++, Makefile, Graph Algorithms. Engineered a graph-based urban transit optimizer supporting shortest distance, minimum stops, and fastest travel time queries. Built interactive route maps with user-friendly visualization for route comparison. Applied Dijkstra's and related algorithms on real-world-style network graphs.
- **Maze-Solver** : BFS & DFS Visualiser, JavaScript. Implemented BFS and DFS path finding on grid-based mazes with animated, step-by-step visualization. Clearly demonstrates algorithmic trade-offs: BFS guarantees shortest path; DFS illustrates depth-first backtracking.

Website, Portfolio and Profiles

github.com/Prasiddhi16

References

DR. SUSHIL SHRESTHA

Associate Professor, Kathmandu University - Mentor

✉ susil@ku.edu.np

SUMAN SHRESTHA

Lecturer, Coordinator, Kathmandu University - Mentor

✉ shrestha.suman@ku.edu.np